

A Later Counterexample to the Hermite Lattice-Density Conjecture in arXiv:0901.4386

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1 Question and answer

Abreu's paper *Sampling and interpolation in Bargmann–Fock spaces of polyanalytic functions*, arXiv:0901.4386, displays the following conjecture: if a lattice $\Lambda \subset \mathbb{R}^2$ makes $\mathcal{G}(h_n, \Lambda)$ a Gabor frame, then

$$D(\Lambda) > n.$$

Here the source defines the density of $\Lambda = A\mathbb{Z}^2$ by $D(\Lambda) = |\det A|^{-1}$. The conjecture has a full negative answer in later literature.

Counterexample. Let

$$\Lambda = \frac{1}{\sqrt{3}}\mathbb{Z}^2.$$

Faulhuber, Shafkulovska, and Zlotnikov prove in Proposition 3.8 of *On the Frame Property of Hermite Functions and Exploration of their Frame Sets*, J. Fourier Anal. Appl. 31:21 (2025), that

$$\mathcal{G}(h_9, \Lambda)$$

is a Gabor frame for $L^2(\mathbb{R})$. On the other hand, the matrix defining this lattice is $A = 3^{-1/2}I_2$, and therefore

$$D(\Lambda) = |\det(3^{-1/2}I_2)|^{-1} = \left|\frac{1}{3}\right|^{-1} = 3.$$

Consequently $\mathcal{G}(h_9, \Lambda)$ is a frame while $D(\Lambda) = 3 \not> 9$. This contradicts the displayed conjecture. $\square$

2 Indexing audit

The source has a one-step indexing inconsistency. Immediately before the displayed conjecture it states the sufficient condition $D(\Lambda) > n$ for $\mathcal{G}(h_{n-1}, \Lambda)$, whereas the conjecture itself uses h_n with the same threshold. The counterexample is insensitive to this discrepancy. On the shifted reading, put $n = 10$, so that $h_{n-1} = h_9$. The conjectured necessity becomes $D(\Lambda) > 10$, which the same density-3 frame also contradicts. It likewise contradicts the usual “safety region is sharp” interpretation $D(\Lambda) > m + 1$ for h_m .

3 Relation to the existing run record

An earlier packet in this run, attached to arXiv:0804.4613, combines a frame equivalence with Theorem 1.1 of the same 2025 paper. It records square-lattice frames at the boundary density $n + 1$ for every $n \geq 4$. That result refutes a strict boundary condition. Proposition 3.8 gives a much stronger separation for the literal conjecture in arXiv:0901.4386: order 9 already admits a square-lattice frame at density 3.

This packet is a literature-implied answer rather than an original proof. It does not classify all lattices in the frame set of h_9 , nor does it answer the source's separate broad questions about Haar-window superframes and vector-valued coherent states.

4 References

- L. D. Abreu, *Sampling and interpolation in Bargmann–Fock spaces of polyanalytic functions*, Appl. Comput. Harmon. Anal. 29 (2010), 287–302; [arXiv:0901.4386](#).
- M. Faulhuber, I. Shafkulovska, and I. Zlotnikov, *On the Frame Property of Hermite Functions and Exploration of their Frame Sets*, J. Fourier Anal. Appl. 31:21 (2025), [doi:10.1007/s00041-025-10153-y](#).